E.G.S. PILLAY ENGINEERING COLLEGE (Autonomous)

Approved by AICTE, New Delhi

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Accredited by NBA T1 (B.E. – CIVIL, CSE, ECE, EEE, MECH & B.Tech – IT)

NAGAPATTINAM–611002

B.E. ELECTRONICS AND COMMUNICATION ENGINEERING

R2023 –THIRD YEAR CURRICULUM

SEMESTER VI

COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS		
							CA	ES	TOTAL
THEORY COURSES									
2302EC601	Antennas and Wave Propagation	PCC	3	0	0	3	40	60	100
2302EC602	Embedded System Design and IoT	PCC	3	0	0	3	40	60	100
2302EC603	Wireless Communication	PCC	3	0	3	4.5	50	50	100
	Professional Elective – II	PEC	3	0	0	3	40	60	100
	Professional Elective – III	PEC	3	0	0	3	40	60	100
	Open Elective – II	OEC	3	0	0	3	40	60	100
PRACTICAL COURSES									
2302EC651	Embedded System Design and IoT Laboratory	PCC	0	0	3	1.5	60	40	100
2302EC652	Mini Project	EEC	0	0	2	1	50	50	100
OTHER COURSES									
	Mandatory Course - II	MC	3	0	0	0	100	--	100
2304GE601	Professional Development Course - IV	EEC	0	0	2	1	100	--	100
2301LS601	Life Skills - VI	LS	--	--	--	--	--	--	--
TOTAL						**23**	**560**	**440**	**1000**

PROFESSIONAL ELECTIVE COURSES

PROFESSIONAL ELECTIVE COURSES

Vertical I	Vertical II	Vertical III	Vertical IV	Vertical V	Vertical VI
Bio Medical Technologies	**Signal Processing**	**Semiconductor Chip Design and Testing**	**High Speed Communications**	**Space Technologies**	**Sensor Technologies and IoT**
Wearable Devices	Advanced Digital Signal Processing	Analog IC Design	4G/5G Communication Networks	Avionics Systems	Fundamentals of Nano-Electronics
Therapeutic Equipment	DSP Architecture and Programming	Low Power IC Design	Advanced Wireless Communication Techniques	Satellite Communication	IoT Based System Design
Human Assist Devices	Digital Image Processing	Mixed Signal IC Design Testing	Massive MIMO Networks	Positioning and Navigation Systems	IoT Processors
Medical Imaging Systems	Speech Processing	Validation and Testing Technology	Optical Communication & Networks	Radar Technologies	Industrial IoT
Body Area Networks	Optical Signal Processing	VLSI Testing and Design For Testability	Software Defined Networks	Remote Sensing	MEMS Design
Brain Computer Interface and Applications	Computer Vision	Wide Bandgap Devices	Wireless BroadBand Networks	Rocketry and Space Mechanics	Wireless Sensor Network Design

PROFESSIONAL ELECTIVE – II

S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC002	Therapeutic Equipment	PEC	3	0	0	3	3
2.	2303EC008	DSP Architecture and Programming	PEC	3	0	0	3	3
3.	2303EC014	Low Power IC Design	PEC	3	0	0	3	3
4.	2303EC020	Advanced Wireless Communication Techniques	PEC	3	0	0	3	3
5.	2303EC026	Satellite Communication	PEC	3	0	0	3	3
6.	2303EC032	IoT Based System Design	PEC	3	0	0	3	3

PROFESSIONAL ELECTIVE – III

S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC003	Human Assist Devices	PEC	3	0	0	3	3
2.	2303EC009	Digital Image Processing	PEC	3	0	0	3	3
3.	2303EC015	Mixed Signal IC Design Testing	PEC	3	0	0	3	3
4.	2303EC021	Massive MIMO Networks	PEC	3	0	0	3	3
5.	2303EC027	Positioning and Navigation Systems	PEC	3	0	0	3	3
6.	2303EC033	IoT Processors	PEC	3	0	0	3	3

OPEN ELECTIVE – II

S.NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC043	High Speed Networks	OEC	3	0	0	3	3
2.	2303EC044	Digital Image Processing	OEC	3	0	0	3	3
3.	2303EC045	Microprocessors and Microcontrollers	OEC	3	0	0	3	3
4.	2303EC046	Wearable Devices	OEC	3	0	0	3	3
5.	2303EC047	Satellite Communication	OEC	3	0	0	3	3
6.	2303EC048	Space Science	OEC	3	0	0	3	3

MANDATORY COURSES – II

Course Code	Course Name
2301MC601	Introduction to Women and Gender Studies
2301MC602	Elements of Literature
2301MC603	Film Appreciation
2301MC604	Disaster Risk Reduction and Management
2301MC605	Well-Being with Traditional Practices-Yoga, Ayurveda and Siddha
2301MC606	History of Science and Technology in India
2301MC607	Political and Economic Thought for a Humane Society
2301MC608	State, Nation Building and Politics in India
2301MC609	Industrial Safety
2301MC610	Stress Management by Yoga
2301MC611	Constitution of India
2301MC612	Essence of Indian Traditional Knowledge
2301MC613	Mental Health & Wellbeing
2301MC614	Universal Human Values – Understanding Harmony And Ethical Human Conduct

2302EC601	ANTENNAS AND WAVE PROPAGATION	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC303 - Electromagnetic Fields

COURSE OBJECTIVES:

- To introduce the fundamental principles of antenna theory and various types of antennas.
- To determine the radiation field of different wire antennas and analyze its parameters.
- To gain knowledge about Antenna arrays.
- To learn the fundamentals of Propagation of Radio waves.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Summarize the fundamentals of antennas in antenna theory.
CO2:	Examine the characteristics of wire antennas, Antenna arrays and establish their mathematical relations.
CO3:	Explore radiation characteristics of Broadband and Smart antennas.
CO4:	Explore radiation characteristics of Aperture antennas and Explain the Radiation Pattern, Gain and Directivity measurements of the antenna.
CO5:	Investigate the attenuation, reflection and refractive characteristics of the Ground wave, Space wave and Sky wave propagation.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2						1		1
CO2	3	3	2	2					1		1
CO3	3	3	2						1		1
CO4	3	3	2						1		1
CO5	3	2	2						1		1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		2
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I	**ANTENNA FUNDAMENTALS**	**9 Hours**

Introduction to antenna Parameters - Radiation Pattern, Radiation intensity, Beam solid angle, Gain, Directive gain, Power gain, Directivity, Beam Width, Band Width, Reciprocity principle, Effective length, Effective area, Relation between gain, effective length and radiation resistance, Friis Transmission formula, Antenna Field Zones, Polarization, Self and mutual impedances of antennas.

MODULE II	**WIRE ANTENNAS AND ANTENNA ARRAYS**	**9 Hours**

Concept of vector potential - Retarded vector potential - Fields associated with alternating current element -

Power radiated and radiation resistance of alternating current element. Radiation from half-wave dipole and quarter-wave monopole, Radiation resistance of half wave dipole and quarter wave monopole - Impedance of Folded dipole. Antenna Arrays: Broadside and End fire array - Expression for electric field from two element arrays – N element linear array - Pattern multiplication- Binomial array.		
MODULE III	**BROADBAND AND SMART ANTENNAS**	**9 Hours**
Loop antennas, Helical antennas, Yagi-Uda antenna, Log periodic antenna, Microstrip antenna. Smart antennas - UWB antennas, RFID Antennas.		
MODULE IV	**APERTURE ANTENNAS AND ANTENNA MEASUREMENTS**	**9 Hours**
Radiation from an elemental area of a plane wave (Huygen's Source), Radiation from a rectangular aperture treated as an array of Huygen's sources, Babinet's principle, Slot antenna- Relation between dipole and slot impedances, Horn antenna – Types, Parabolic reflector antenna and its feed systems. Antenna Measurements: Radiation Pattern Measurement, Gain and Directivity Measurements.		
MODULE V	**RADIO WAVE PROPAGATION**	**9 Hours**
Modes of propagation, Structure of atmosphere, Ground wave propagation - Space wave propagation- Duct propagation - Sky wave propagation - Mechanism of refraction - Refractive index- Critical frequency - Skip distance - Maximum usable frequency – Fading - Multi hop propagation.		
		TOTAL: 45 HOURS

REFERENCES:

1. *John D.Kraus and RonalatoryMarhefka, "Antennas for all Applications", Tata McGraw Hill, Third Edition, 2010.*
2. *Constantine. A Ballanis, "Antenna Theory: Analysis and Design", John Wiley & Sons, 2nd Edition, 2016.*
3. *K.D. Prasad, "Antenna and Wave Propagation",Sathyaprakasan Tech India Publications- New Delhi-2011.*
4. *Robert. E. Collin, "Antennas and Radio Propagation", McGraw-Hill, 2004.*
5. *A.R. Harish, M. Sachidanada, "Antennas and Wave propagation", Oxford University Press, 2011.*
6. *S. Drabowitch, "Modern Antennas", Springer Publications, 2nd Edition, 2007.*

2302EC602	EMBEDDED SYSTEM DESIGN AND IOT	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC401 - Analog Integrated Circuits
2. 2302EC402 - Analog Communication
3. 2302EC501 - Digital Communication
4. 2302EC502 - Networks and Security

COURSE OBJECTIVES:

To know about the architecture and features of 8051.
To study the design process of an embedded system.
To understand the real –time processing in an embedded system.
To learn the architecture and design flow of IoT.
To develop an IoT based system

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the architecture and features of 8051.
CO2:	Build a model of an embedded system.
CO3:	List the concepts of real time operating systems.
CO4:	Make the architecture and protocols of IoT.
CO5:	Design an IoT based system for any application.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	1				1		1
CO2	3	3	2	2	1			1	1	1	1
CO3	3	3	2	2	1			1	1		1
CO4	3	3	2	2	1			1			2
CO5	3	3	3	3	1			1		1	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	2
CO2	3	2
CO3	3	2
CO4	3	3
CO5	3	3

COURSE CONTENTS:

MODULE I	**MICROCONTROLLERS**	**9 Hours**

Microcontrollers for an Embedded System – 8051 – Architecture – Addressing Modes – Instruction Set– Program and Data Memory–Interrupts – Timers/Counters –Architecture of Advanced Microcontrollers - Arduino UNO, LPC 2148 Microcontroller.

MODULE II	**EMBEDDED SYSTEMS**	**9 Hours**

Embedded System Design Process – Model Train Controller – ARM Processor – Instruction Set Preliminaries – CPU – Programming Input and Output – Supervisor Mode – Exceptions and Trap –Models for programs – Assembly, Linking and Loading – Compilation Techniques – Program Level Performance Analysis.		
MODULE III	**PROCESSES AND OPERATING SYSTEMS**	**9 Hours**
Structure of a real – time system – Task Assignment and Scheduling – Multiple Tasks and Multiple Processes – Priority based scheduling – Inter-process Communication Mechanisms –Distributed Embedded Systems – MPSoCs and Shared Memory Multiprocessors – Design Example –Engine Control Unit and Video Accelerator.		
MODULE IV	**IOT ARHITECTURE AND PROTOCOLS**	**9 Hours**
Internet – of – Things – Physical Design, Logical Design – IoT Enabling Technologies – Domain Specific IoTs – IoT and M2M – IoT Platform Design – Methodology – IoT– IoT Protocols -MQTT, XMPP, Modbus, CAN BUS and BAC Net.		
MODULE V	**IOT SYSTEM DESIGN**	**9 Hours**
Basic building blocks of an IoT device – Raspberry Pi – Board – Linux on Raspberry Pi – Case Studies: RealTime Automation, Smart Cities, Environment and Agriculture.		
TOTAL: 45 HOURS		
Experiments are based on the Industry curriculum. Syllabus drawn from the Hitakey Technology and EmCog Solutions Pvt. Ltd.		
REFERENCES:		
1. Mohammed Ali Mazidi, Janice Gillispie Mazidi, Rolin D.McKinlay, The 8051 Microcontroller and Embedded Systems Using Assembly and C, Second Edition, Pearson Education, 2008.		
2. Marilyn Wolf, Computers as Components – Principles of Embedded Computing System Design, Third Edition, Morgan Kaufmann, 2012.		
3. Arshdeep Bahga, Vijay Madisetti, Internet – of- Things – Hands on Approach, Universities Press, 2015.		
4. Mayur Ramgir, Internet–of Things, Architecture, Implementation and Security, First Edition, Pearson Education, 2020.		
5. Jane.W.S.Liu, Real –Time Systems, Pearson Education, 2003.		
6. Lyla B .Das, Embedded Systems: An Integrated Approach, Pearson Education, 2013.		

2302EC603	WIRELESS COMMUNICATION	L	T	P	C
		3	0	3	4.5

PREREQUISITE:

1. 2302EC402 - Analog Communication
2. 2302EC501 - Digital Communication

COURSE OBJECTIVES:

- To study and understand the concepts and design of a cellular system.
- To study and understand mobile radio propagation and various digital modulation techniques.
- To understand the concepts of multiple access techniques and wireless networks.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the concept and design of a cellular system using simulation tools
CO2:	Explain different types of the mobile radio propagation
CO3:	Classify various digital modulation techniques.
CO4:	Demonstrate the concepts of multiple access techniques
CO5:	Analyze the different generations of Wireless Networks.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2		2						1
CO2	3	2	2		2						1
CO3	3	2	2		2						1
CO4	3	2	2		2						1
CO5	3	2	2		2						1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I	THE CELLULAR CONCEPT-SYSTEM DESIGN FUNDAMENTALS	9 Hours

Introduction-Frequency Reuse-Channel Assignment Strategies-**Handoff Strategies:** Prioritizing Handoffs, Practical Handoff Considerations. **Interference And System Capacity:** Co-Channel Interference and System Capacity-Channel Planning for Wireless Systems, Adjacent Channel Interference, Power Control for Reducing Interference, Trunking and Grade of Service. **Improving Coverage and Capacity in Cellular Systems:** Cell Splitting, Sectoring.

MODULE II	MOBILE RADIO PROPAGATION	9 Hours

Large Scale Path Loss: Introduction to Radio Wave Propagation - Free Space Propagation Model – **Three Basic Propagation Mechanism:** Reflection – Brewster Angle- Diffraction Scattering. **Small Scale Fading and Multipath:** Small Scale Multipath Propagation, Factors Influencing Small-Scale Fading, Doppler Shift, Coherence Bandwidth, Doppler Spread and Coherence Time.

MODULE III	MODULATION TECHNIQUES AND OFDM	9 Hours

Digital Modulation – An Overview: Factors That Influence The Choice of Digital Modulation, Linear Modulation Techniques: Minimum Shift Keying (MSK), Gaussian Minimum Shift Keying(GMSK), **Orthogonal Frequency Division Multiplexing –** Introduction, Transmitter, Receiver, Applications		
MODULE IV	**MULTIPLE ACCESS TECHNIQUES**	**9 Hours**
Introduction: Introduction To Multiple Access- Frequency Division Multiple Access (FDMA) - Time Division Multiple Access (TDMA) - Spread Spectrum Multiple Access - Code Division Multiple Access (CDMA) - Space Division Multiple Access (SDMA).		
MODULE V	**WIRELESS NETWORKING**	**9 Hours**
Introduction: Difference Between Wireless and Fixed Telephone Networks, The Public Switched Telephone Network (PSTN), Development of Wireless Networks: First Generation Wireless Networks, Second Generation Wireless Networks, Third Generation Wireless Networks, Fixed Network Transmission Hierarchy, TrafficRouting inWireless Networks: Circuit Switching, Packet Switching.		
		TOTAL: 45 HOURS

LIST OF EXPERIMENTS:

1. Modeling of wireless communication systems using MATLAB (Two ray channel and Okumura–Hata model).
2. Modeling and simulation of Multipath fading channel using MATLAB.
3. Design, analyze and test Wireless standards and evaluate the performance measurements such as BER, PER, BLER, throughput, capacity, ACLR, EVM for 4G and 5G using MATLAB.
4. Modulation: Spread Spectrum – DSSS Modulation & Demodulation using MATLAB.
5. Wireless Channel equalization: Zero-Forcing Equalizer (ZFE), MMSE Equalizer (MMSEE), Adaptive Equalizer (ADE), Decision Feedback Equalizer (DFE) using MATLAB.
6. Modeling and Simulation of TDMA, FDMA and CDMA for wireless communication using MATLAB.
7. Virtual Lab.

TOTAL: 45 HOURS

REFERENCES:

1.Rappaport,T.S.,-Wireless communications", Pearson Education, Second Edition, 2010.

2.Andrea Goldsmith, Wireless Communication, Cambridge University Press, 2011.

3.Van Nee, R. and Ramji Prasad, OFDM for wireless multimedia communications, Artech House, 2000.

4.David Tse and PramodViswanath, Fundamentals of Wireless Communication, Cambridge University Press, 2005.

5.UpenaDalal, Wireless Communication, Oxford University Press, 2009.

6.William Stallings, Wireless Communication and Networks,Pearson Education, Second Edition, 2002.

2302EC651	EMBEDDED SYSTEM DESIGN AND IOT LABORATORY	L	T	P	C
		0	0	3	1.5

PREREQUISITE:

1. 2302EC451 - Analog Integrated Circuits Lab
2. 2302EC551 - Communication Systems Lab

COURSE OBJECTIVES:

- To perform the arithmetic and logical operations in 8051.
- To introduce interfacing, I/o Devices to the Processor.
- To introduce the evolution of the Internet of Things (IoT).
- To apply the concept of the Internet of Things in real-world scenario.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Understand the arithmetic and logical operation in 8051 microcontroller.
CO2:	Learn the ARM processor with Interfacing of ADC and DAC.
CO3:	Learn the ARM processor with Interfacing of LED and LCD.
CO4:	Design an Embedded based system with real time application.
CO5:	Design an IoT based system with real time application.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2						1
CO2	3	3	2	2	2						1
CO3	3	3	2	2	2						1
CO4	3	3	2	2	2						2
CO5	3	3	3	3	3						2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	2
CO2	3	3
CO3	3	3
CO4	3	3
CO5	3	3

LIST OF EXPERIMENTS:

Experiments using 8051.

1. Programming Arithmetic and Logical Operations in 8051.
2. Generation of Square waveform using 8051.

Experiments Using ARM

3. Interfacing of ADC and DAC
4. Blinking of LED.
5. Interfacing with LCD.
6. Interfacing Keyboard.
7. Interfacing Stepper Motor.

Mini Projects for Embedded and IOT
8. Design an automatic night light using an LDR to control an LED.
9. Smart Greenhouse Monitoring System.
10. Object Detection and Alerts..
TOTAL: 45 HOURS
Experiments are based on the Industry curriculum. Syllabus drawn from the Hitakey Technology and EmCog Solutions Pvt. Ltd.
REFERENCES:
1. *B.Arunkumar, "Embedded and Iot design Laboratory Manual", 2021.*
2. *https://www.studocu.com/in/document/mount-litera-zee-school/computer-science-sl/cs3691-embedded-systems-and-iot-lab-manual/88752899.*
3. *https://www.studocu.com/in/document/vel-tech-rangarajan-dr-sagunthala-rd-institute-of-science-and-technology/embedded-system-design/es-iot-lab-manual/84437016*
4. *https://www.slideshare.net/slideshow/embedded-systems-and-iot-lab-manual-for-enginnering-students/267160236*

PROFESSIONAL ELECTIVE – II

S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC002	Therapeutic Equipment	PEC	3	0	0	3	3
2.	2303EC008	DSP Architecture and Programming	PEC	3	0	0	3	3
3.	2303EC014	Low Power IC Design	PEC	3	0	0	3	3
4.	2303EC020	Advanced Wireless Communication Techniques	PEC	3	0	0	3	3
5.	2303EC026	Satellite Communication	PEC	3	0	0	3	3
6.	2303EC032	IoT Based System Design	PEC	3	0	0	3	3

2303EC002	THERAPEUTIC EQUIPMENT	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC201 - Electron Devices

COURSE OBJECTIVES:

- To learn the principles of cardiac assist devices.
- To understand the need and use of extracorporeal devices, and the use of lasers inmedicine.
- To enable the students to gain knowledge on the working of therapeutic clinical equipment.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Choose the suitable therapeutic devices for ailments related to cardiology, pulmonology, neurology, etc.
CO2:	Summarize the principles of body care equipment.
CO3:	Explain the operation of dental care equipment.
CO4:	Examine the different types of therapies for suitable applications.
CO5:	Illustrate the application of lasers in biomedical applications.

COs Vs POs MAPPING:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							3
CO2	3	3	3	2							3
CO3	3	3	3	2							3
CO4	3	3	3	2							3
CO5	3	3	3	2							3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	
CO2	2	
CO3	2	
CO4	2	
CO5	2	

COURSE CONTENTS:

MODULE I	CARDIAC AND RESPIRATORY THERAPY EQUIPMENT	9 Hours

Cardiac Pacemaker: Internal and External Pacemaker– Programmable pacemakers. Cardiac Defibrillators: AC and DC Defibrillator- Internal and External Defibrillators - Protection Circuit, Defibrillator analyzers. Cardiac ablation catheter.

Types of Ventilators – Pressure, Volume, and Time controlled. Basic principles of electromechanical, pneumatic and electronic ventilators, Patient Cycle Ventilators, Ventilator testing. Humidifiers, Nebulizers, Inhalators.

MODULE II	BIOMECHANICAL THERAPEUTIC EQUIPMENT	9 Hours

Electrodiagnosis, Therapeutic radiation, Electrotherapy, Electrodes, Stimulators for Nerve and Muscle, Functional Electrical Stimulation. Peripheral nerve stimulator, ultrasonic stimulators, Stimulators for pain and relief - Inferential Therapy Unit, TENS. GAIT Assessment and Therapy. Continuous Passive Motion unit, Cervical / Lumber Traction Machine - Traction Table.

MODULE III	**BODY CARE EQUIPMENT**	**9 Hours**
Skin Treatment: Ultrasonic spot remove, vacuum therapy unit, Skin tightening, Wrinkle Reduction, Facial and Rejuvenation. Laser hair therapy machine. Body Slimmer/Shaper – Deep Heat Therapy, Massager, Fitness – Treadmill, Bike.		
MODULE IV	**DENTAL CARE EQUIPMENT**	**9 Hours**
Dental Chair - Dental Hand pieces and Accessories: Evolution of rotary equipment, Low-speed hand piece, High-speed hand piece, Hand piece maintenance. Vacuum and Pneumatic techniques: Vacuum techniques, Oral evacuation systems, Vacuum pump, Pneumatic techniques, Dental compressor. Decontamination Unit and constant fumigation unit. Dental Radiography: Dental X-ray Machine.		
MODULE V	**HEAT & PHOTON THERAPY EQUIPMENT**	**9 Hours**
High frequency heat therapy, Principle, Short wave diathermy, Microwave diathermy, Ultrasonic therapy, Lithotripsy. Therapeutic UV and IR Lamps. Basic principles of Biomedical LASERS: Applications of lasers in medicine, CO2laser, He-Ne laser, Nd-YAG and Ruby laser.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Leslie Cromwell, Fred. J. Weibell & Erich. A.Pfeiffer, "Biomedical Instrumentation and Measurements", Second Edition, Prentice Hall Inc. 2000.*
2. *John Low & Ann Reed, "Electrotherapy Explained, Principles and Practice", Second Edition, Butterworth Heinemann Ltd. 2000.*
3. *Joseph. J. Carr, John Michael Brown, "Introduction to Biomedical Equipment Technology", Prentice Hall and Technology, 2008.*

2303EC008	DSP ARCHITECTURE AND PROGRAMMING	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC301 - Signals & Systems
2. 2302EC404 - Digital Signal Processing

COURSE OBJECTIVES:

- Study the architecture of programmable DSP processors
- Learn to implement various standard DSP algorithms in DSP Processors
- Use the Programmable DSP Processors to build real-time DSP systems

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the architectural features of DSP Processors.
CO2:	Sketch the organization of TMS320C54xx DSP processors
CO3:	Build a solutions using TMS320C6x DSP Processor
CO4:	Implement the DSP Algorithms
CO5:	Outline the applications of DSP Processors.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3								3
CO2	3	3	2								2
CO3	3	3	3								2
CO4	3	3	3								2
CO5	3	3	3								2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I	ARCHITECTURES FOR PROGRAMMABLE DSP PROCESSORS	9 Hours

Basic Architectural features, DSP Computational building blocks, Bus architecture and memory, Data addressing capabilities, Address generation Unit, Programmability and program execution, Speed issues, Features for external interfacing

MODULE II	RF TMS320C5X PROGRAMMABLE DSP PROCESSOR	9 Hours

Architecture of TMS320C54xx DSP processors, Addressing modes – Assembly language Instructions - Memory space, interrupts, and pipeline operation of TMS320C54xx DSP Processor, On-Chip peripherals, Block Diagram of TMS320C54xx DSP starter kit

MODULE III	TMS320C6X PROGRAMMABLE DSP PROCESSOR	9 Hours

Commercial TI DSP processors, Architecture of TMS320C6x DSP Processor, Linear and Circular addressing modes, TMS320C6x Instruction Set, Assembler directives, Linear Assembly, Interrupts, Multichannel buffered serial ports, Block diagram of TMS320C67xx DSP Starter Kit and Support Tools		
MODULE IV	**IMPLEMENTATION OF DSP ALGORITHMS**	**9 Hours**
DSP Development system, On-chip, and On-board peripherals of C54xx and C67xx DSP development boards, Code Composer Studio (CCS) and support files, Implementation of Conventional FIR, IIR, and Adaptive filters in TMS320C54xx/TMS320C67xx DSP processors for real-time DSP applications, Implementation of FFT algorithm for frequency analysis in real-time.		
MODULE V	**APPLICATIONS OF DSP PROCESSORS**	**9 Hours**
Voice scrambling using filtering and modulation, Voice detection and reverse playback, Audio effects, Graphic Equalizer, Adaptive noise cancellation, DTMF signal detection, Speech thesis using LPC, Automatic speaker recognition		
		TOTAL: 45 HOURS

REFERENCES:

1. *Avtar Singh and S. Srinivasan, Digital Signal Processing – Implementations using DSP Microprocessors with Examples from TMS320C54xx, Cengage Learning India Private Limited, Delhi 2012*
2. *RulphChassaing and Donald Reay, Digital Signal Processing and Applications with the TMS320C6713 and TMS320C6416 DSK, Second Edition, Wiley India (P) Ltd, New Delhi, 2008*
3. *B.Venkataramani and M.Bhaskar, "Digital Signal Processors – Architecture, Programming and Applications", Tata McGraw – Hill Publishing Company Limited. New Delhi, 2003.*
4. *TMS320C5416/6713 DSK user manual at https://www.ti.com*

2303EC014	LOW POWER IC DESIGN	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC302 Digital Electronics- Detailed knowledge about Combinational Circuits, Sequential circuits, memories and programmable logics.
2. 2302EC401 Analog Integrated Circuits- Concepts of analog IC's and Operational amplifiers.

COURSE OBJECTIVES:

- Getting familiar with various IC technology.
- Learn MOS theory and testing
- Learn CMOS circuit theory and testing
- Getting expertise on CMOS characterization.
- Explore circuit and device level testing methods

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Acquire the fundamental knowledge of Low power circuit design.
CO2:	Attain the knowledge of architectural approaches.
CO3:	Analyze and design Low-Voltage Low-Power combinational circuits.
CO4:	Design of Low-Voltage Low-Power Memories
CO5:	Design and develop Low Power, Low Voltage Circuits

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	1	2				2	2
CO2	3	3	3	2	2	2				2	2
CO3	3	3	3	2	2	2				2	2
CO4	3	3	3	2	2	2				2	2
CO5	3	3	3	2	2	2				2	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	
CO2	1	
CO3	1	
CO4	1	
CO5	2	

COURSE CONTENTS:

MODULE I	FUNDAMENTALS OF LOW POWER CIRCUITS	9 Hours

Need for Low Power Circuit Design, Sources of Power Dissipation – Switching Power Dissipation, Short Circuit Power Dissipation, Leakage Power Dissipation, Glitching Power Dissipation, Short Channel Effects –Drain Induced Barrier Lowering and Punch Through, Surface Scattering, Velocity Saturation, Impact Ionization, Hot Electron Effect

MODULE II	LOW-POWER DESIGN APPROACHES	9 Hours

Low-Power Design through Voltage Scaling: VTCMOS circuits, MTCMOS circuits, Architectural Level Approach –Pipelining and Parallel Processing Approaches. Switched Capacitance Minimization

Approaches: System Level Measures, Circuit Level Measures, Mask level Measures.		
MODULE III	**LOW-VOLTAGE LOW-POWER ADDERS**	**9 Hours**
Introduction, Standard Adder Cells, CMOS Adder's Architectures – Ripple Carry Adders, Carry Look-Ahead Adders, Carry Select Adders, Carry Save Adders, LowVoltage Low Power Design Techniques – Trends of Technology and Power Supply Voltage, LowVoltage Low-Power Logic Styles.		
MODULE IV	**LOW-VOLTAGE LOW-POWER MULTIPLIERS**	**9 Hours**
Introduction, Overview of Multiplication, Types of Multiplier Architectures, Braun Multiplier, Baugh-Wooley Multiplier, Booth Multiplier, Introduction to Wallace Tree Multiplier		
MODULE V	**LOW-VOLTAGE LOW-POWER MEMORIES**	**9 Hours**
Basics of ROM, Low-Power ROM Technology, Future Trend and Development of ROMs, Basics of SRAM, Memory Cell, Precharge and Equalization Circuit, LowPower SRAM Technologies, Basics of DRAM, Self-Refres Circuit, Future Trend and Development of DRAM.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Sung-Mo Kang, Yusuf Leblebici, "CMOS Digital Integrated Circuits – Analysis and Design", TMH, 2011.*
2. *Kiat-Seng Yeo, Kaushik Roy, "Low-Voltage, Low-Power VLSI Subsystems", TMH Professional Engineering, 2004.*
3. *Ming-BO Lin, "Introduction to VLSI Systems: A Logic, Circuit and System Perspective", CRCPress, 2012.*
4. *Anantha Chandrakasan, "Low Power CMOS Design", IEEE Press, /Wiley International, 1998*
5. *Kaushik Roy, Sharat C. Prasad, "Low Power CMOS VLSI Circuit Design", John Wiley, &Sons, 2000.*
6. *Gary K. Yeap, "Practical Low Power Digital VLSI Design", Kluwer Academic Press, 2002*
7. *Bellamour, M. I. Elamasri, "Low Power CMOS VLSI Circuit Design", A Kluwer Academic Press, 1995.*
8. *Siva G. Narendran, Anatha Chandrakasan, "Leakage in Nanometer CMOS Technologies", Springer, 2005*

2303EC020	ADVANCED WIRELESS COMMUNICATION TECHNIQUES	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC402 - Analog Communication
2. 2302EC501 – Digital Communication

COURSE OBJECTIVES:

- To understand the evolving paradigm of cooperative communication
- To understand concepts related to green wireless communication
- To enable the student to understand the different power saving strategies and energy efficient signal, system and network design.
- To expose the student to the energy saving techniques adopted in existing wireless components
- To provide understanding on protocols and networks related to green future wireless communication technologies.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Implement the design aspects of cooperative communication.
CO2:	Build the design aspects of green wireless communication.
CO3:	Evolve the new techniques in wireless communication.
CO4:	Demonstrate the feasibility of using mathematical models using simulation tools.
CO5:	Determine the impact of the green engineering solutions in a global, economic, environmental and societal context.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	1						2
CO2	3	3	2	2	1						1
CO3	3	2	2	2	1	1					1
CO4	3	3	2	2	2						1
CO5	3	2	2	2	1	2					2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I	**COOPERATIVE COMMUNICATIONS AND GREEN CONCEPTS**	**9 Hours**

Network architectures and research issues in cooperative cellular wireless networks; Cooperative communications in OFDM and MIMO cellular relay networks: issues and approaches; Fundamental trade-offs on the design of green radio networks, Green modulation and coding schemes.		
MODULE II	**COOPERATIVE TECHNIQUES**	**9 Hours**
Cooperative techniques for energy efficiency, Cooperative base station techniques for cellular wireless networks; Turbo base stations; Antenna architectures for cooperation; Cooperative communications in 3GPP LTE-Advanced, Partial information relaying and Coordinated multi-point transmission in LT E-Advanced.		
MODULE III	**RELAY-BASED COOPERATIVE CELLULAR NETWORKS**	**9 Hours**
Distributed space-time block codes ; Collaborative relaying in downlink cellular systems ; Radio resource optimization; Adaptive resource allocation ; Cross-layer scheduling design for cooperative wireless two-way relay networks; Network coding in relay-based networks.		
MODULE IV	**GREEN RADIO NETWORKS**	**9 Hours**
Base Station Power-Management Techniques- Opportunistic spectrum and load management, Energy-saving techniques in cellular wireless base stations, Power-management for base stations in smart grid environment, Cooperative multi cell processing techniques for energy-efficient cellular wireless communications.		
MODULE V	**ACCESS TECHNIQUES FOR GREEN RADIO NETWORKS**	**9 Hours**
Cross-layer design of adaptive packet scheduling for green radio networks; Energy-efficient relaying for cooperative cellular wireless networks; Energy performance in TDD-CDMA multihop cellular networks; Resource allocation for green communication in relay-based cellular networks; Green Radio Test-Beds and Standardization Activities.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Ekram Hossain, Dong In Kim, Vijay K. Bhargava, "Cooperative Cellular Wireless Networks", Cambridge University Press, 2011.*
2. *Ekram Hossain, Vijay K. Bhargava(Editor), Gerhard P. Fettweis (Editor), "Green RadioCommunication Networks", Cambridge University Press, 2012.*
3. *F. Richard Yu, Yu, Zhang and Victor C. M. Leung "Green Communications and Networking", CRC press, 2012.*
4. *Ramjee Prasad and Shingo Ohmori, Dina Simunic, "Towards Green ICT", River Publishers, 2010.*
5. *Jinsong Wu, Sundeep Rangan and Honggang Zhang, "Green Communications: Theoretical Fundamentals, Algorithms and Applications", CRC Press, 2012.*

2303EC026	SATELLITE COMMUNICATION	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC402 - Analog Communication
2. 2302EC501 – Digital Communication
3. 2302EC603 - Wireless Communication

COURSE OBJECTIVES:

- Understand the basics of satellite orbits.
- Understand the satellite segment and earth segment.
- Understand Link Power budget calculation.
- Understand the various satellite access and coding technology.
- Understand the applications of satellite.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Identify the satellite orbits.
CO2:	Analyze the satellite subsystems.
CO3:	Evaluate the satellite link power budget.
CO4:	Identify access technology for satellite.
CO5:	Design various satellite applications.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	1	1	1					1
CO2	3	3	3	1	1	1					1
CO3	3	3	3	2	1	1					1
CO4	3	3	3	2	1	1					1
CO5	3	3	3	2	1	1					1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		2
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I	**SATELLITE ORBITS**	**9 Hours**

Kepler's Laws, Newton's law, orbital parameters, orbital perturbations, station keeping, Geo stationary and Non-Geo-stationary orbits – Look Angle Determination- Limits of visibility – eclipseSub satellite point –Sun transit outage-Launching Procedures - launch vehicles and propulsion.

MODULE II	SPACE SEGMENT	**9 Hours**

Spacecraft Technology- Structure, Primary power, Attitude and Orbit control, Thermal control and Propulsion, communication Payload and supporting subsystems, Telemetry, Tracking and command-Transponders Antenna Subsystem.		
MODULE III	**SATELLITE LINK DESIGN**	**9 Hours**
Basic link analysis, Uplink and Downlink Design equation, Free space loss-Atmospheric effects, Ionospheric scintillation, Rain induced attenuation and interference, system noise temperature, Link Design with and without frequency reuse		
MODULE IV	**SATELLITE ACCESS AND CODING TECHNIQUES**	**9 Hours**
Modulation and Multiplexing: Voice, Data, Video, Analog – digital transmission system, Digital video Broadcast, multiple access: FDMA, TDMA, CDMA, PAMA and DAMA Assignment Methods, compression – encryption, Coding Schemes		
MODULE V	**SATELLITE APPLICATIONS**	**9 Hours**
INTELSAT Series, INSAT, VSAT, Mobile satellite services: GSM, GPS, LEO, MEO, Satellite Navigational System. GPS-Position Location Principles, Differential GPS, Direct Broadcast satellites (DBS/DTH).		
		TOTAL: 45 HOURS

REFERENCES:

1. *Dennis Roddy, "Satellite Communication", 4th Edition, Mc Graw Hill International, 2017.*
2. *Timothy Pratt, Charles, W.Bostain, Jeremy E.Allnutt,"SatelliteCommunication",3rd Edition, Wiley Publications,2021.*
3. *Tri T. Ha, "Digital Satellite Communications", 2nd edition, Mc Graw Hill education, 2017*
4. *Wilbur L.Pritchard, Hendri G. Suyderhoud, Robert A. Nelson, "Satellite Communications Systems Engineering", 2nd edition , Prentice Hall/Pearson , 2013*
5. *M.Richharia, "Satellite Communication Systems-Design Principles", Macmillan, 1999.*
6. *Brian Ackroyd, "World Satellite Communication and earth station Design", BSP professional Books, 1990*
7. *Bruce R. Elbert, "The Satellite Communication Applications", Hand Book, Artech House Bostan London, 2003.*

2303EC032	IOT BASED SYSTEM DESIGN	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC301 - Signals and Systems
2. 2302EC301 - Digital Electronics
3. 2302EC402 - Analog Communication
4. 2302EC501 – Digital Communication

COURSE OBJECTIVES:

To understand the basics of IoT.
To get knowledge about the various services provided by IoT.
To familiarize themselves with various communication techniques and networking.
To know the implementation of IoT with different tools.
To understand the various applications in IoT.

COURSE OUTCOMES:

Upon completion of this course, the students will be able to

CO1:	Articulate the main concepts, key technologies, strength and limitations of IoT.
CO2:	Identify the architecture, infrastructure models of IoT.
CO3:	Analyze the networking and how the sensors are communicated in IoT.
CO4:	Analyze and design different models for IoT implementation.
CO5:	Identify and design the new models for market strategic interaction.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	1			2	2	2	2
CO2	3	3	2	2	1			2	2	2	2
CO3	3	3	3	2	1			2	2	2	2
CO4	3	3	2	2	1			2	2	2	2
CO5	3	3	3	3	1			2	2	2	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	3
CO2	3	3
CO3	3	2
CO4	3	3
CO5	3	2

COURSE CONTENTS:

MODULE I	INTRODUCTION TO INTERNET OF THINGS	9 Hours

Rise of the machines – Evolution of IoT – Web 3.0 view of IoT – Definition and characteristics of IoT – IoT Enabling Technologies – IoT Architecture -– Fog, Edge and Cloud in IoT – Functional blocks of an IoT ecosystem – Sensors, Actuators, Smart Objects and Connecting Smart Objects - IoT levels and deployment templates – A panaromic view of IoT applications.		
MODULE II	**MIDDLEWARE AND PROTOCOLS OF IOT**	**9 Hours**
Middleware technologies for IoT system (IoT Ecosystem Overview – Horizontal Architecture Approach for IoT Systems – SOA based IoT Middleware) Middleware architecture of RFID, WSN, SCADA, M2M – Interoperability challenges of IoT - Protocols for RFID, WSN, SCADA, M2M - Zigbee, KNX, BAC Net, MODBUS - Challenges Introduced by 5G in IoT Middleware(Technological Requirements of 5G Systems - Perspectives and a Middleware Approach Toward 5G (COM PaaS Middleware) – Resource management in IoT.		
MODULE III	**COMMUNICATION AND NETWORKING**	**9 Hours**
IoT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.15.4g, 802.15.4e, 1901.2a, 802.11ah and LoRaWAN – Network Layer: IP versions, Constrained Nodes and Constrained Networks – Optimizing IP for IoT: From 6LoWPAN to 6Lo, Routing over Low Power and Lossy Networks – Application Transport Methods: Supervisory Control and Data Acquisition –Application Layer Protocols: CoAP and MQTT- Data aggregation & dissemination.		
MODULE IV	**IOT IMPLEMENTATION TOOLS**	**9 Hours**
Introduction to Python, Introduction to different IoT tools, Developing applications through IoT tools,- Developing sensor based application through embedded system platform- Implementing IoT concepts with python, Implementation of IoT with Raspberry Pi.		
MODULE V	**APPLICATIONS AND CASE STUDIES**	**9 Hours**
Home automations - Smart cities – Environment – Energy – Retail – Logistics – Agriculture – Industry - Health and life style – Case study.		
		TOTAL: 45 HOURS

REFERENCES

1. *Honbo Zhou, "Internet of Things in the cloud:A middleware perspective", CRC press, 2012.*
2. *Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-on Approach)", VPT, 1st Edition, 2014.*
3. *Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press, 2017.*
4. *Constandinos X. Mavromoustakis, George Mastorakis, Jordi Mongay Batalla, "Internet of Things (IoT) in 5G Mobile Technologies" Springer International Publishing Switzerland 2016.*
5. *Dieter Uckelmann, Mark Harrison, Florian Michahelles, "Architecting the Internet of Things", Springer-Verlag Berlin Heidelberg, 2011.*

PROFESSIONAL ELECTIVE – III

S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC003	Human Assist Devices	PEC	3	0	0	3	3
2.	2303EC009	Digital Image Processing	PEC	3	0	0	3	3
3.	2303EC015	Mixed Signal IC Design Testing	PEC	3	0	0	3	3
4.	2303EC021	Massive MIMO Networks	PEC	3	0	0	3	3
5.	2303EC027	Positioning and Navigation Systems	PEC	3	0	0	3	3
6.	2303EC033	IoT Processors	PEC	3	0	0	3	3

2303EC003	HUMAN ASSIST DEVICES	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC201 - Electron Devices

COURSE OBJECTIVES:

- To study the role and importance of machines that takes over the functions of the heartand lungs,
- To study various mechanical techniques that help a non-functioning heart.
- To learn the functioning of the unit which does the clearance of urea from the blood
- To understand the tests to assess the hearing loss and development of electronic devicesto compensate for the loss.
- To study about recent techniques used in modern clinical applications

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the principles and construction of artificial heart
CO2:	Choose various mechanical techniques that improve therapeutic technology
CO3:	Explain the functioning of the membrane or filter that cleanses the blood.
CO4:	Describe the tests to assess the hearing loss and development of wearable devices for the same.
CO5:	Analyze and research on electrical stimulation and biofeedback techniques in rehabilitationand physiotherapy.

COs Vs POs MAPPING:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							2
CO2	3	3	3	2							2
CO3	3	3	3	2							2
CO4	3	3	3	2							2
CO5	3	3	3	2							2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	2
CO2	3	2
CO3	3	2
CO4	3	2
CO5	3	2

COURSE CONTENTS:

MODULE I	HEART LUNG MACHINE AND ARTIFICIAL HEART	9 Hours

Condition to be satisfied by the H/L System. Different types of Oxygenators, Pumps, Pulsatile and Continuous Types, Monitoring Process, Shunting, The Indication for Cardiac Transplant, Driving Mechanism, Blood Handling System, Functioning and different types of Artificial Heart, Schematic for temporary bypass of left ventricle.

MODULE II	CARDIAC ASSIST DEVICES	9 Hours

Assisted through Respiration, Right and left Ventricular Bypass Pump, Auxiliary ventricle, Open Chest and Closed Chest type, Intra-Aortic Balloon Pumping, Prosthetic Cardiac valves, Principle of External Counter pulsation techniques.		
MODULE III	**ARTIFICIAL KIDNEY**	**9 Hours**
Indication and Principle of Haemodialysis, Membrane, Dialysate, types of filter and membranes, Different types of hemodialyzers, Monitoring Systems, Wearable Artificial Kidney, Implanting Type.		
MODULE IV	**RESPIRATORY AND HEARING AIDS**	**9 Hours**
Ventilator and its types - Intermittent positive pressure, Breathing Apparatus Operating Sequence, Electronic IPPB unit with monitoring for all respiratory parameters. Types of Deafness, Hearing Aids, SISI, masking techniques, wearable devices for hearing correction.		
MODULE V	**RECENT TRENDS**	**9 Hours**
Transcutaneous electrical nerve stimulator, bio-feedback, Diagnostic and point-of-care platforms.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Andreas.F. Von racum, "Hand book of bio material evaluation", Mc-Millan publishers, 1980.*
2. *Gray E Wnek, Gray L Browlin, "Encyclopedia of Biomaterials and Biomedical Engineering"Marcel Dekker Inc New York 2004.*
3. *D.S. Sunder, "Rehabilitation Medicine", 3rd Edition, Jaypee Medical Publication, 2010.*

2303EC009	DIGITAL IMAGE PROCESSING	L	T	P	C
		3	0	0	3

PREREQUISITE:

3. 2302EC301 - Signals & Systems
4. 2302EC404 - Digital Signal Processing

COURSE OBJECTIVES:

- To become familiar with digital image fundamentals
- To get exposed to simple image enhancement techniques in Spatial and Frequency domain
- To learn concepts of degradation function and restoration techniques.
- To study the image segmentation and representation techniques.
- To become familiar with image compression and recognition methods

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Summarize the fundamentals of digital image processing, such as digitization, sampling, quantization, and 2D-transforms.
CO2:	Choose an image using the techniques of smoothing, sharpening and enhancement.
CO3:	Dramatize the restoration concepts and filtering techniques.
CO4:	Distinguish the concept of segmentation, features extraction, compression and recognition methods for color models.
CO5:	Breakdown the image compression concepts.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							3
CO2	3	3	3	2							2
CO3	3	3	2	2							2
CO4	3	3	3	2							2
CO5	3	3	3	3							2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	2
CO2	2	2
CO3	2	1
CO4	2	1
CO5	2	1

COURSE CONTENTS:

MODULE I	DIGITAL IMAGE FUNDAMENTALS	9 Hours

Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels - Color image fundamentals

- RGB, HSI models, Two-dimensional mathematical preliminaries, 2D transforms - DFT, DCT.		
MODULE II	**IMAGE ENHANCEMENT**	**9 Hours**
Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering – Smoothing and Sharpening Spatial Filtering, Frequency Domain: Introduction to Fourier Transform– Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters, Homomorphic filtering, Color image enhancement.		
MODULE III	**IMAGE RESTORATION**	**9 Hours**
Image Restoration - degradation model, Properties, Noise models – Mean Filters – Order Statistics – Adaptive filters – Band reject Filters – Band pass Filters – Notch Filters – Optimum Notch Filtering – Inverse Filtering – Wiener filtering		
MODULE IV	**IMAGE SEGMENTATION**	**9 Hours**
Edge detection, Edge linking via Hough transforms – Thresholding - Region based segmentation – Region growing – Region splitting and merging – Morphological processing- erosion and dilation, Segmentation by morphological watersheds – basic concepts – Dam construction – Water shed segmentation algorithm.		
MODULE V	**IMAGE COMPRESSION AND RECOGNITION**	**9 Hours**
Need for data compression, Huffman, Run Length Encoding, Shift codes, Arithmetic coding, JPEG standard, MPEG. Boundary representation, Boundary description, Fourier Descriptor, Regional Descriptors – Topological feature, Texture - Patterns and Pattern classes - Recognition based on matching.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Rafael C. Gonzalez, Richard E. Woods, 'Digital Image Processing', Pearson, Third Edition, 2010.*
2. *Anil K. Jain, 'Fundamentals of Digital Image Processing', Pearson, 2002.*
3. *Kenneth R. Castleman, 'Digital Image Processing', Pearson, 2006.*
4. *Rafael C. Gonzalez, Richard E. Woods, Steven Eddins, 'Digital Image Processing using MATLAB', Pearson Education, Inc., 2011.*
5. *D.E. Dudgeon and RM. Mersereau, 'Multidimensional Digital Signal Processing', Prentice Hall Professional Technical Reference, 1990.*
6. *William K. Pratt, 'Digital Image Processing', John Wiley, New York, 2002*
7. *Milan Sonka et al 'Image processing, analysis and machine vision', Brookes/Cole, Vikas Publishing House, 2nd edition, 1999.*

2303EC015	MIXED SIGNAL IC DESIGN TESTING	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC401 Analog Integrated Circuits – Knowledge about A/D and D/A converters

COURSE OBJECTIVES:

- To know about mixed-signal devices and the need for testing these devices.
- To study the various techniques for testing.
- To learn about ADC and DAC based testing.
- To understand the Clock and Serial Data Communications Channels
- To study the general purpose measuring devices.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Attain the knowledge of mixed signal circuits.
CO2:	Define the various measurement terminologies.
CO3:	Acquire knowledge of Analog to Digital Converters.
CO4:	Testing of Analog to Digital Converters.
CO5:	Comprehend the attributes of a clock signal.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	1	2		2		2	2
CO2	3	3	2	2	2	2		2		2	2
CO3	3	3	2	2	2	2		2		2	2
CO4	3	3	3	2	2	2		2		2	2
CO5	3	3	3	2	3	2		2		2	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	
CO2	3	
CO3	3	
CO4	1	
CO5	2	

COURSE CONTENTS:

MODULE I	MIXED – SIGNAL TESTING	9 Hours

Common Types of Analog and Mixed- Signal Circuits – Applications of Mixed-Signal Circuits - Post-Silicon Production Flow - Test and Packing – Characterization versus Production Testing - Test and Diagnostic Equipment - Automated Test Equipments – Wafer Probers – Handlers – E-Beam Probers – Focused Ion Beam Equipments – Forced –Temperature

MODULE II	**YIELD, MEASUREMENT ACCURACY, AND TEST TIME**	**9 Hours**
Yield - Measurement Terminology - Repeatability, Bias, and Accuracy - Calibrations and Checkers - Tester Specifications - Reducing Measurement Error with Greater Measurement Time – Guardbands - Effects of Measurement Variability on Test Yield - Effects of Reproducibilty and Process Variation on Yield - Statistical Process Control		
MODULE III	**DAC TESTING**	**9 Hours**
Basics of Data Converters -Principles of DAC and ADC Conversion, Data Formats, Comparison of DACs and ADCs, DAC Failure Mechanisms - Basic DC Tests - Transfer Curve Tests - Dynamic DAC Tests - Tests for Common DAC Applications		
MODULE IV	**ADC TESTING**	**9 Hours**
ADC Testing Versus DAC Testing - ADC Code Edge Measurements - Edge Code Testing Versus Center Code Testing, Step Search and Binary Search Methods, Servo Method, Linear Ramp Histogram Method, Histograms to Code Edge Transfer Curves, Rising Ramps Versus Falling Ramps, Sinusoidal Histogram Method - DC Tests and Transfer Curve Tests - Dynamic ADC Tests - Tests for Common ADC Applications		
MODULE V	**CLOCK AND SERIAL DATA COMMUNICATIONS CHANNEL MEASUREMENT**	**9 Hours**
Synchronous and Asynchronous Communications - Time-Domain Attributes of a Clock Signal - Frequency-Domain Attributes of a Clock Signal - Communicating Serially Over a Channel - Bit Error Rate Measurement - Methods to Speed Up BER Tests in Production - Deterministic Jitter Decomposition - Jitter Transmission Tests.		
		TOTAL: 45 HOURS

REFERENCES:

1. *M.L.Bushnell and V.D.Agrawal, "Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits", Kluwer Academic Publishers, 2002.*
2. *BapirajuVinnakota, "Analog and mixed-signal test", Prentice Hall, 1998.*
3. *Digital and Analogue Instrumentation: Testing and Measurement by Nihal Kularatna*
4. *Gordon W.Roberts, Friedrich Taenzler, Mark Burns, "An Introduction to Mixed-signal IC Test and Measurement" Oxford University Press, Inc.2012.*

2303EC021	MASSIVE MIMO NETWORKS	L	T	P	C
		3	0	0	3

PREREQUISITE:

3. 2302EC402 - Analog Communication
4. 2302EC501 – Digital Communication

COURSE OBJECTIVES:

To gain knowledge about massive MIMO networks.

To understand the massive MIMO propagation channels.

To learn about channel estimation in single cell and multicell massive MIMO systems.

To comprehend the concepts of massive MIMO deployment in the context of single cell and multicell deployment.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Understand and explain massive MIMO networks.
CO2:	Analyze massive MIMO propagation channels and their capacity bounds.
CO3:	Examine channel estimation techniques for single cell system.
CO4:	Analyze channel estimation techniques for multi cell system.
CO5:	Explain the concepts underlining the deployment of single and multicell massive MIMO systems.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	1					1	2
CO2	3	3	2	2	1	1				1	1
CO3	3	2	2	2	1					1	1
CO4	3	3	2	2	1	1				1	1
CO5	3	2	2	2	1	1				1	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		2
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I	MASSIVE MIMO NETWORKS	9 Hours

Definition of Massive MIMO, Correlated Rayleigh Fading, System Model for Uplink and Downlink, Basic Impact of Spatial Channel Correlation, Channel Hardening and Favourable Propagation, Local Scattering Spatial Correlation Model.		
MODULE II	**THE MASSIVE MIMO PROPAGATION CHANNEL**	**9 Hours**
Favorable Propagation and Deterministic Channels-Capacity Upper Bound-Distance from Favorable Propagation-Favorable Propagation and Linear Processing-Singular Values and Favorable Propagation, Favorable Propagation and Random Channels-Independent Rayleigh Fading-Uniformly Random Line-of-Sight (UR-LoS)-Independent Rayleigh Fading versus UR-LoS - Finite-Dimensional Channels		
MODULE III	**SINGLE-CELL SYSTEMS**	**9 Hours**
Uplink Pilots and Channel Estimation - Orthogonal Pilots- De-Spreading of the Received Pilot Signal-MMSE Channel Estimation, Uplink Data Transmission - Zero-Forcing -Maximum-Ratio, Downlink Data Transmission-Linear Precoding-Zero-Forcing-Maximum-Ratio, DiscussionInterpretation of the Effective SINR Expressions-Implications for Power Control-Scaling Laws and Upper Bounds on the SINR - Near-Optimality of Linear Processing when M >> K - Net Spectral Efficiency - Limiting Factors: Number of Antennas and Mobility		
MODULE IV	**MULTI-CELL SYSTEMS**	
Uplink Pilots and Channel Estimation, Uplink Data Transmission - Zero-Forcing - Maximum-Ratio, Downlink Data Transmission - Zero-Forcing - Maximum-Ratio, Discussion - Asymptotic Limits with Infinite Numbers of Base Station Antennas - The Effects of Pilot Contamination - Non-Synchronous Pilot Interference.		
MODULE V	**CASE STUDIES**	**9 Hours**
Single-Cell Deployment Example: Fixed Broadband Access in Rural Area, Multi-Cell Deployment: Preliminaries and Algorithms, Multi-Cell Deployment Examples: Mobile Access - Dense Urban 178 Scenario - Suburban Scenario - Minimum Per-Terminal Throughput Performance -Additional Observations - Comparison of Power Control Policies.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Thomas L. Marzetta, Erik G. Larsson, Hong Yang, Hien Quoc Ngo, "Fundamentals of Massive MIMO", Cambridge University Press 2016. (UNITS II-V)*
2. *Emil Björnson, Jakob Hoydis and Luca Sanguinetti (2017), "Massive MIMO Networks: Spectral, Energy, and Hardware Efficiency", Foundations and Trends, Now, 2017. (UNIT I)*
3. *Long Zhao, Hui Zhao, Kan Zheng, "Wei Xiang Massive MIMO in 5G Networks: Selected Applications", Springer 2018.*
4. *Leibo Liu, Guiqiang Peng, Shaojun Wei, "Massive MIMO Detection Algorithm and VLSI Architecture", Springer 2019*
5. *Shahid Mumtaz, Jonathan Rodriguez, Linglong Dai, "mmWave Massive MIMO A Paradigm for 5G", Elsevier, 2017*

2303EC027	POSITIONING AND NAVIGATION SYSTEMS	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC402 - Analog Communication
2. 2302EC501 – Digital Communication

COURSE OBJECTIVES:

To explain the fundamentals of navigation systems.
To understand the inertial navigation systems
To acquire knowledge on radio navigation.
To have an overview of global positioning systems
To learn the hybrid navigation systems.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Describe the advanced concepts of Positioning and Navigation systems and exposure on various Navigation systems.
CO2:	Describe about Gyroscopes and accelerometers and Inertial Navigation systems and its types and Mechanisation.
CO3:	Explain the different Radio Navigation aids and its usage for civil and military aircrafts and Satellites.
CO4:	Explain the Satellite Navigation – GPS and its usage in aircraft and spacecraft applications.
CO5:	Deploy these skills effectively in the analysis and understanding of hybrid navigation systems and Relative navigation in a spacecraft.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	1						1	3
CO2	3	3	3	2						1	2
CO3	3	3	3	2						1	2
CO4	3	3	3	2						1	2
CO5	3	3	3	2						1	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		2
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I	**NAVIGATION CONCEPTS**	**9 Hours**
Fundamentals of navigation systems and Position Fixing – Categories of navigation – Geometric concepts of Navigation – The Earth in inertial space - Different Coordinate Systems – Coordinate Transformation - Euler angle formulations - Direction cosine matrices formulation – Quaternion formulation.		
MODULE II	**INERTIAL NAVIGATION SYSTEMS**	**9 Hours**
Inertial sensors - Gyroscopes -Types - Mechanical - Electromechanical-Optical Gyro - Ring Laser gyro-Fiber optic gyro- Accelerometers – Pendulous type – Force Balance type – MEMs – Basic Principles of Inertial Navigation – Types - Platform and Strap down - Mechanization INS system - Rate Corrections - Acceleration errors – Schuler Tuning.		
MODULE III	**RADIO NAVIGATION & AIR TRAFFIC MANAGEMENT**	**9 Hours**
Different types of radio navigation- ADF, VOR, DME, TACAN, VORTAC - Doppler – Hyperbolic Navigations – Air Traffic Management – RADAR Surveillance - Airborne Collision Avoidance Systems.		
MODULE IV	**GLOBAL POSITIONING SYSTEM**	**9 Hours**
Overview of GPS: Basic concept, system architecture - GPS Signals Signal structure, anti-spoofing (AS), selective availability - GPS for position and velocity determination - GPS aided Geo-augmented navigation (GAGAN) architecture -GPS error sources-clock error - ionospheric error - tropospheric error, multipath, ionospheric error estimation using dual frequency GPS receiver.		
MODULE V	**HYBRID NAVIGATION & RELATIVE NAVIGATION SYSTEMS**	**9 Hours**
Hybrid Navigation - Introduction to Kalman filtering – Case Studies -Integration of GPS and INS using Kalman Filter - Relative Navigation – fundamentals – Equations of Relative Motion for circular orbits (Clohessy_Wiltshire Equations) - Sensors for Rendezvous Navigation - Relative positioning - Point positioning and differential positioning - Differential GPS (DGPS) and Space based Augmentation system (SBAS)- Concepts - Relative GPS -Formation Flying - Figure of Merit (FOM).		
		TOTAL: 45 HOURS

REFERENCES:

1. *Myron Kyton, Walfred Fried, "Avionics Navigation Systems", John Wiley & Sons, 2nd Edition, 1997.*
2. *Nagaraja, N.S, "Elements of Electronic Navigation", Tata McGraw-Hill Pub. Co., New Delhi, 2nd Edition, 1975.*
3. *George M Siouris, "Aerospace Avionics System; A Modern Synthesis", Academic Press Inc., 1993.*
4. *Albert Helfrick, 'Practical Aircraft Electronic Systems", Prentice Hall Education, Career &Technology, 1995.*
5. *Albert D. Helfrick, "Modern Aviation Electronics", 2nd Edition, Prentice Hall Career & Technology, 1994.*
6. *Paul. D. Groves, "Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems", Artech House, 2013.*
7. *Maxwell Noton, "Spacecraft navigation and guidance", Springer (London, New York), 1998.*
8. *Myron Kyton, Walfred Fried, "Avionics Navigation Systems", John Wiley & Sons, 2nd Edition, 1997.*

2303EC033	IOT PROCESSORS	L	T	P	C
		3	0	0	3

PREREQUISITE:

5. 2302EC301 - Signals and Systems
6. 2302EC301 - Digital Electronics
7. 2302EC402 - Analog Communication
8. 2302EC501 – Digital Communication

COURSE OBJECTIVES:

- Learn the architecture and features of ARM.
- Study the exception handling and interrupts in CORTEX M3.
- Program the CORTEX M3.
- Learn the architecture of STM 32L15XXX ARM CORTEX M3/M4 microcontroller.
- Understand the concepts of System – On – Chip (SoC).

COURSE OUTCOMES:

On successful completion of this course, the student will be able to

CO1:	Explain the architecture and features of ARM.
CO2:	Summarize the concepts of exception handling.
CO3:	Write a program using ARM CORTEX M3/M4.
CO4:	Examine the architecture of STM32L15XXX ARM CORTEX M3/M4.
CO5:	Design a SoC for any application.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2				1		1	2
CO2	3	3	3	3				1		1	2
CO3	3	3	3	3				1		1	2
CO4	3	3	2	2				1		1	2
CO5	3	3	2	2				1		1	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	
CO2	3	
CO3	3	
CO4	2	
CO5	2	

COURSE CONTENTS:

MODULE I	**OVERVIEW OF ARM AND CORTEX-M3**	**9 Hours**

ARM Architecture – Versions, Instruction Set Development, Thumb 2 and Instruction Set Architecture, Cortex M3 Basics: Registers, Stack Pointer, Link Register, Program Counter, Special Registers, Operation Mode, Exceptions and Interrupts, Vector Tables, Stack Memory Operations, Reset Sequence , CORTEX M3 Instruction Sets: Assembly Basics, Instruction List, Instruction Descriptions, CORTEX M3 – Implementation Overview: Pipeline, Block Diagram. Bus Interfaces, I – Code Bus, D – Code Bus, System Bus- External PPB and DAP Bus.

MODULE II	**CORTEX EXCEPTION HANDLING AND INTERRUPTS**	**9 Hours**

Exception Types, Priority, Vector Tables, Interrupt Inputs and Pending behaviour, Fault Exceptions, Supervisor Call and Pendable Service Call, NVIC: Nested Vector Interrupt Controller, Overview, Basic Interrupts, SYSTICK Time, Interrupt Behaviourm Interrupt/Exception Sequences, Exception Exits, Nested Interrupts, Tail – Chaining Interrupts, Late Arrivals and Interrupt Latency.

MODULE III	**CORTEX M3/M4 PROGRAMMING**	**9 Hours**

Cortex M3/M4 Programming: Overview, Typical Development Flow, Using C, CMSIS Using Assembly, Exception Programming Using Interrupts, Exception/Interrupt Handlers, Software Interrupts, Vector Table Relocation, Memory Protection Unit and other CORTEX M3 Features, MPU Registers, Setting up the MPU, Power Management, Multiprocessor Configuration.

MODULE IV	**STM32L15XXX ARMCORTEX M3/M4 MICROCONTROLLER AND DEBUGGING TOOLS**	**9 Hours**

STM32L15XXX ARM CORTEX M3/M4 Microcontroller: Memory and Bus Architecture, Power Control, Reset and Clock Control, STM32L15XXX Peripherals: GPIOs, System Configuration Controller, NVIC, ADC, Comparators, GP Timers, USART Development and Debugging Tools: Software and Hardware tools like Cross Assembler Compiler, Debugger, Simulator, In – Circuit Emulator (ICE), Logic Analyzer.

MODULE V	**INTRODUCTION TO SYSTEM – ON – CHIP**	**9 Hours**

System Architecture: An Overview, Components of the System Processors, Memories and Interconnects, Processor Architectures, Memory and Addressing, System Level Interconnection – An Approach for SOC Design – Chip basics – Cycle Time – Die Area – Power and Cost – Area, Power and Time Trade – Offs in Processor Design – Reliability and Configurability – SOC Design Approach – Application Studies – AES, 3D Graphics Processor. Image Compression and Video Compression.

TOTAL: 45 HOURS

REFERENCES

1. *Joseph Yiu, The Definitive Guide to the ARM CORTEX M3/M4, Second Edition, Elsevier, 2010.*
2. *Andrew N Sloss, Dominic Symes, Chris Wright, ARM System Developers Guide Designing and Optimising System Software, Elsevier, 2006.*
3. *Michael J Flynn and Wayne Luk, Computer System Design, System On Chip, Wiley India 2011.*
4. *Steve Furber, ARM System – on – Chip Architecture, 2nd Edition, Pearson, 2015.*
5. *CORTEX M Series ARM Reference Manual*
6. *CORTEX M3 Technical Reference Manual*
7. *STM32L152XX ARM CORTEX M3 Microcontroller Reference Manual 5/97.*

OPEN ELECTIVES – II

S.NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC043	High Speed Networks	OEC	3	0	0	3	3
2.	2303EC044	Digital Image Processing	OEC	3	0	0	3	3
3.	2303EC045	Microprocessors and Microcontrollers	OEC	3	0	0	3	3
4.	2303EC046	Wearable Devices	OEC	3	0	0	3	3
5.	2303EC047	Satellite Communication	OEC	3	0	0	3	3
6.	2303EC048	Space Science	OEC	3	0	0	3	3

2303EC043	HIGH SPEED NETWORKS	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

To tell important concepts of networking.
To study traffic management and tunneling protocols for security.
To learn about protocols for better service.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Know basics of Networks
CO2:	Understand congestion concepts
CO3:	Examine advanced networking techniques
CO4:	Illustrate Traffic modelling concepts
CO5:	Review Quality of service protocols

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3								1
CO2	3	3	3								1
CO3	3	3	3								1
CO4	3	3	3								1
CO5	3	3	3								1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I	INTRODUCTION	9 Hours

Frame Relay Networks – Asynchronous transfer mode – ATM Protocol Architecture, ATM logical Connection, ATM Cell – ATM Service Categories – AAL, High Speed LANs: Fast Ethernet, Gigabit Ethernet, Fiber Channel – Wireless LANs: applications, requirements – Architecture of 802.11

MODULE II	CONGESTION AND TRAFFIC MANAGEMENT	9 Hours

Queuing Analysis- Queuing Models – Single Server Queues – Effects of Congestion – Congestion Control – Traffic Management – Congestion Control in Packet Switching Networks – Frame Relay Congestion Control.

MODULE III	ADVANCED NETWORKS CONCEPTS	9 Hours

VPN-Remote-Access VPN, site-to-site VPN, Tunneling to PPP, Security in VPN. MPLS-operation, Routing, Tunneling and use of FEC, Traffic Engineering, MPLS based VPN, overlay networks-P2P connections.		
MODULE IV	**TRAFFIC MODELLING**	**9 Hours**
Little's theorem, Need for modeling, Poisson modeling and its failure, Non-poisson models, Network performance evaluation.		
MODULE V	**PROTOCOLS FOR QOS SUPPORT**	**9 Hours**
RSVP – Goals & Characteristics, Data Flow, RSVP operations, Protocol Mechanisms – Multiprotocol Label Switching – Operations, Label Stacking, Protocol details – RTP – Protocol Architecture, Data Transfer Protocol, RTCP.		
		TOTAL: 45 HOURS

REFERENCES:

1. *J.F. Kurose&K.W. Ross, "Computer Networking- A top down approach featuring the internet", Pearson 2nd edition, 2003.*
2. *Walrand.J. Varatya, "High performance communication network", Morgan Kauffman – Harcourt Asia Pvt. Ltd, 2nd Edition, 2000.3.*
3. *Aunurag kumar, D.Manjunath, Joykuri, "Communication Networking", Morgan Kaufmann Publishers, First Edition. 2004.*
4. *Hersent Gurle & petit, "IP Telephony, packet Pored Multimedia communication Systems", Pearson Education, 2003.*
5. *Larryl. Peterson & Bruce S.David, "Computer Networks: A System Approach", 1996.*

2303EC044	DIGITAL IMAGE PROCESSING	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

1.To become familiar with digital image fundamentals
2.To get exposed to simple image enhancement techniques in Spatial and Frequency domain
3.To learn concepts of degradation function and restoration techniques.
4.To study the image segmentation and representation techniques.
5.To become familiar with image compression and recognition methods

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the fundamentals of image processing.
CO2:	Choose the image enhancement technique for the image applications.
CO3:	Classify the image compression techniques.
CO4:	Solve the problem using image segmentation concepts.
CO5:	Identify the properties and characteristics of image and recognize it.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3								3
CO2	3	3	3								3
CO3	3	3	3								3
CO4	3	3	3								3
CO5	3	3	3								3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I	DIGITAL IMAGE FUNDAMENTALS	9 Hours

Elements of Visual perception. Image sensing and Acquisition. Imaging in different bands. Digital Image Representation. Relationship between pixels. Image transformations: 2D-DFT, DCT, DST, Hadamard, Walsh, Hotelling transformation, 2D-Wavelet transformation, Wavelet packets.

MODULE II	IMAGE ENHANCEMENT	9 Hours

Image Enhancements in spatial domain and Frequency domain. Image Restoration techniques. Color Image processing.		
MODULE III	**IMAGE COMPRESSION**	**9 Hours**
Error free compression: Variable length coding, LZW, Bit-plane coding, Lossless predictive coding Lossy compression: Lossy predictive coding, transform coding, wavelet coding. Image compression standards, CCITT, JPEG, JPEG 2000, Video compression standards.		
MODULE IV	**IMAGE SEGMENTATION**	**9 Hours**
Summary of morphological operations in Binary and Gray Images. Image segmentation: Point, Line and Edge segmentation. Edge linking and Boundary detection. Segmentation using thresholding, Region based segmentation. Segmentation by morphological watersheds. Use of motion in segmentation.		
MODULE V	**IMAGE IDENTIFICATION AND RECOGNITION**	**9 Hours**
Feature Extraction from the Image: Boundary descriptors, Regional descriptors, Relational descriptors.		
		TOTAL: 45 HOURS

REFERENCES:

8. *Rafael C.Gonzalez, Richard E.Woods, 'Digital Image Processing', Pearson, Third Edition, 2010.*
9. *Anil K.Jain, 'Fundamentals of Digital Image Processing', Pearson, 2002.*
10. *Kenneth R. Castleman, 'Digital Image Processing', Pearson, 2006.*
11. *Rafael C. Gonzalez, Richard E. Woods, Steven Eddins, 'Digital Image Processing using MATLAB', Pearson Education, Inc., 2011.*
12. *D.E. Dudgeon and RM. Mersereau, 'Multidimensional Digital Signal Processing', Prentice Hall Professional Technical Reference, 1990.*
13. *William K. Pratt, 'Digital Image Processing', John Wiley, New York, 2002*
14. *Milan Sonka et al 'Image processing, analysis and machine vision', Brookes/Cole, Vikas Publishing House, 2nd edition, 1999.*

2303EC045	MICROPROCESSOR AND MICROCONTROLLER	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To teach the architecture and functions of 8085 and 8086 Microprocessors.
- To impart the concepts of 8051 microcontroller.
- To convey aspects of I/O and Memory Interfacing circuits
- To study and understand applications of microcontroller
- To develop skills in simple program writing in assembly languages.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Construct hardware, software and programming concepts of Microprocessor.
CO2:	Summarize architecture, instructions and addressing modes of 8086 Microprocessor.
CO3:	Summarize architecture, instructions and addressing modes of 8086 Microprocessor.
CO4:	Illustrate interfacing of Serial, parallel, Keyboard, Display with Microcontroller.
CO5:	Use the programming concepts to write assembly language programs.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2								2
CO2	3	2	2								2
CO3	3	3	2								2
CO4	3	3	2								2
CO5	3	3	2								2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	
CO2	3	
CO3	3	
CO4	3	
CO5	3	

COURSE CONTENTS:

MODULE I	INTRODUCTION TO MICROPROCESSORS	9 Hours

Evolution of Microprocessors - 8-Bit Processor - 8085 Architecture, Register Organization, Instruction Sets, Timing Diagram, Addressing Modes, Interrupts, Interrupt Service Routines- Assembly Language Programming Using 8085.

MODULE II	8086 MICROPROCESSORS	9 Hours

Introduction to 8086 – Microprocessor architecture – Addressing modes - Instruction set and assembler directives – Assembly language programming – Modular Programming - Linking and Relocation - Stacks - Procedures – Macros – Interrupts and interrupt service routines - 8086 signals .

MODULE III	MICROCONTROLLERS	9 Hours

Architecture of 8051 – Special Function Registers (SFRs) - I/O Pins Ports and Circuits - Instruction set - Addressing modes - Assembly language programming.		
MODULE IV	**I/O INTERFACING**	**9 Hours**
Memory Interfacing and I/O interfacing - Parallel communication interface – Serial communication interface – D/A and A/D Interface - Timer – Keyboard /display controller – Interrupt controller – DMA controller		
MODULE V	**ADVANCED MICROPROCESSOR & MICROCONTROLLER**	**9 Hours**
Advanced coprocessor Architectures- 286, 486, Pentium -RISC Processors- RISC Vs CISC, RISC properties and evolution- ARM Processor fundamentals, ARM Architecture – Register, CPSR, Pipeline, exceptions and interrupts interrupt vector table		
		TOTAL: 45HOURS

REFERENCES:

1. *R. S. Gaonkar, "Microprocessor Architecture: Programming and Applications with the 8085/8080A", Penram International Publishing, Third Edition.*
2. *D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface" Morgan Kaufman Publishers, Fourth Edition, 2011.*
3. *Douglas Hall, "The Microprocessors and its Interfacing", Tata McGraw Hill, Third Edition, 2012*
4. *Kenneth J. Ayala, "The 8051 Microcontroller: Architecture Programming & Applications", Penram International Publishing, Second Edition, 1996*
5. *Yu-Cheng Liu, Glenn A. Gibson, "Microcomputer Systems: The 8086 / 8088 Family - Architecture, Programming and Design", Second Edition, Prentice Hall of India, 2011*
6. *Mohamed Ali Mazidi, Janice Gillispie Mazidi, Rolin McKinlay, "The 8051 Microcontroller and Embedded Systems: Using Assembly and C", Second Edition, Pearson education, 2011*
7. *Doughlas V. Hall, "Microprocessors and Interfacing, Programming and Hardware", Second Edition, TMH, 2012. B.E. Electronics and Communication Engineering | E.G.S. Pillay Engineering College (Autonomous) | Regulations2019 Approved in IV Academic Council Meeting held on 25-05-2019 Page 10 of 15*
8. *John P. Hayes, "Computer Architecture and Organization", Third illustrated Edition, Tata McGraw Hill, 2007.*

2303EC046	WEARABLE DEVICES	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To know the hardware requirement of wearable systems
- To understand the communication and security aspects in the wearable devices
- To know the applications of wearable devices in the field of medicine
- To analyze the technical aspects of wireless health systems
- To apply the inculcated knowledge in wearable sensors for developing simple medical applications

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Differentiate the sensors that can be used for wearable systems
CO2:	Process the signals picked by the wearable sensors
CO3:	Utilize different sources of energy to be used for wearable systems
CO4:	Analyze the technical aspects of wireless health systems
CO5:	Apply the wearable sensors into novel medical applications

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							3
CO2	3	3	3	2							3
CO3	3	3	3	2							3
CO4	3	3	3	2							3
CO5	3	3	3	2							3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	
CO2	2	
CO3	2	
CO4	2	
CO5	2	

COURSE CONTENTS:

MODULE I	INTRODUCTION TO WEARABLE SYSTEMS AND SENSORS	9 Hours

Wearable Systems- Introduction, Need for Wearable Systems, Drawbacks of Conventional Systems for Wearable Monitoring, Applications of Wearable Systems, Types of Wearable Systems, Components of wearable Systems. Sensors for wearable systems-Inertia movement sensors, Respiration activity sensor Inductive plethysmography, Impedance plethysmography, pneumography, Wearable ground reaction force sensor.

MODULE II	SIGNAL PROCESSING FOR WEARABLEDEVICES	9 Hours

Wearability issues -physical shape and placement of sensor, technical challenges - sensor design, signal acquisition, Constraint on sampling frequency for reduced energy consumption, light weight signal processing, Rejection of irrelevant information, Data mining.		
MODULE III	**ENERGY HARVESTING FOR WEARABLE DEVICES**	**9 Hours**
Solar cell, Vibration based, Thermal based human body as a heat source for power generation, Hybrid thermoelectric photovoltaic energy harvests, Thermopiles.		
MODULE IV	**WIRELESS HEALTH SYSTEMS**	**9 Hours**
Need for wireless monitoring, ISO/OSI layer architecture, Definition of Body area network, BAN and Healthcare, Technical Challenges- System security and reliability, BAN Architecture – Introduction, Wireless communication techniques.		
MODULE V	**APPLICATIONS OF WEARABLE SYSTEMS**	**9 Hours**
Medical Diagnostics, Medical Monitoring-Patients with chronic disease, Hospital patients, Elderly patients, neural recording, Gait analysis, Sports Medicine.		
		TOTAL: 45 HOURS
REFERENCES:		
1. *Sandeep K.S, Gupta, Tridib Mukherjee and Krishna Kumar Venkatasubramanian, Body Area Networks Safety, Security, and Sustainability, Cambridge University Press, 2013*		
2. *Guang-Zhong Yang, Body Sensor Networks, Springer, 2006.*		
3. *Andreas Lymberis, Danilo de Rossi ,'Wearable eHealth systems for Personalised Health Management - State of the art and future challenges ' IOS press, The Netherlands, 2004.*		

2303EC047	SATELLITE COMMUNICATION	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

To explain about satellites, their operation, types, navigation and launch

To discuss about satellites access and antenna systems

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Discuss orbital mechanics and launch methodologies.
CO2:	Recognize various space subsystems.
CO3:	Explain different subsystems of earth segment
CO4:	Discuss the Principles of navigation and launching
CO5:	Demonstrate various Satellite Applications

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3								1
CO2	3	2	2								1
CO3	3	3	3								1
CO4	3	3	2								1
CO5	3	2	3								1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I	**SATELLITE ORBITS**	**9 Hours**

Introduction - Spectrum allocations for satellite systems - Kepler‘s Laws - orbital parameters - orbital perturbations – Type of orbits - Geo stationary orbits – Look angle determination - Limits of visibility – Eclipse - Sub satellite point – Sun transit outage.

MODULE II	**SPACE AND EARTH SEGMENT**	**9 Hours**

Spacecraft technology – Structure - Power supply - Attitude and Station keeping, Orbit control - Thermal control - Communication subsystems - Telemetry, Tracking and Command – Transponders - Antenna subsystem, Earth station Technology - Receive only home TV systems - MATV – CATV.

MODULE III	**SATELLITE ACCESS**	**9 Hours**

Modulation and Multiplexing-Voice, Data, Video, Analog – Digital transmission system - Digital video Broadcast - Multiple Accesses: (FDMA, TDMA, CDMA, SDMA - assignment methods) -Spread Spectrum Communication.		
MODULE IV	**SATELLITE NAVIGATIONAL SYSTEM**	**9 Hours**
GPS Principle of operation, Position Location Determination, Principle of GPS Receiver and Applications - Launching procedures - Launch vehicles and propulsion.		
MODULE V	**SATELLITE APPLICATIONS**	**9 Hours**
Satellite Mobile services – VSAT- Radarsat - Direct Broadcast Satellites (DBS) - Direct to Home Broadcast (DTH) - Digital Audio Broadcast (DAB) – World Space Services, Business TV (BTV) – GRAMSAT - Specialized services: E mail, Video conferencing, Internet - INTELSAT Series - INSAT – INMARSAT.		
		TOTAL: 45 HOURS
REFERENCES:		

1. *Dennis Roddy, "Satellite Communication", McGraw Hill International, 4th Edition, 2006.*
2. *Wilbur L.Pritchard, Hendri G. Suyderhoud, Robert A. Nelson, "Satellite Communication Systems Engineering", Prentice Hall/Pearson, 2007.*
3. *N.Agarwal, "Design of Geosynchronous Space Craft", Prentice Hall, 1986.*
4. *Bruce R. Elbert, "The Satellite Communication Applications", Hand Book, Artech House Bostan London, 1997.*
5. *Tri T. Ha, "Digital Satellite Communication", 2nd Edition, 1990.*
6. *N.S.Nagaraja, "Elements electronic navigation system", 2nd Edition, Tata McGraw Hill, 2000.*

2303EC048	SPACE SCIENCE	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To outline the space environment and their effects.
- To extend the origin of universe and development.
- To classify the galaxies and their evolution.
- To interpret the variable stars in the galaxies.
- To explain theory of formation of our solar system.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Obtain a broad, basic knowledge of the space sciences.
CO2:	Explain the scientific concepts such as evolution by means of natural selection, age of the Earth and solar system and the Big-Bang.
CO3:	Describe the main features and formation theories of the various types of observed galaxies, in particular the Milky Way.
CO4:	Explain stellar evolution, including red giants, supernovas, neutron stars, pulsars, whitedwarfs and black holes, using evidence and presently accepted theories;
CO5:	Describe the presently accepted formation theories of the solar system based upon observational and physical constraints;

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2								1
CO2	3	3	2								1
CO3	3	3	2								1
CO4	3	3	2								1
CO5	3	3	2								1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		1
CO2		1
CO3		1
CO4		1
CO5		1

COURSE CONTENTS:

MODULE I	INTRODUCTION	9 Hours

Introduction to space science and applications – historical development – Space Environment- Vacuum and its Effects, Plasma & Radiation Environments and their Effects, Debris Environment and its Effects - Newton's Law of gravitation – Fundamental Physical Principles.

MODULE II	ORIGIN OF UNIVERSE	9 Hours

Early history of the universe – Big-Bang and Hubble expansion model of the universe – cosmic microwave background radiation – dark matter and dark energy.		
MODULE III	**GALAXIES**	**9 Hours**
Galaxies, their evolution and origin – active galaxies and quasars – Galactic rotation – Stellar populations – galactic magnetic field and cosmic rays.		
MODULE IV	**STARS**	**9 Hours**
Stellar spectra and structure – stellar evolution – Nucleo-synthesis and formation of elements – Classification of stars – Harvard classification system – Hertsprung-Russel diagram – Luminosityof star – variable stars – composite stars (white dwarfs, Neutron stars, black hole, star clusters, supernova and binary stars) – Chandrasekhar limit.		
MODULE V	**SOLAR SYSTEM**	**9 Hours**
Nebular theory of formation of our Solar System – Solar wind and nuclear reaction as the source ofenergy – Sun and Planets: Brief description about shape size – period of rotation about axis and period of revolution – distance of planets from sun – Bode's law – Kepler's Laws of planetary motion – Newton's deductions from Kepler's Laws – correction of Kepler's third law – determinationof mass of earth – determination of mass of planets with respect to earth – Brief description of Asteroids – Satellites and Comets.		
		TOTAL: 45 HOURS
REFERENCES:		

1. *Hess W., "Introduction to Space Science", Gordon & Breach Science Pub; Revised Ed., 1968.*
2. *Krishnaswami K. S., "Astrophysics: A modern Perspective", New Age International, 2006.*
3. *Arnab Rai Choudhuri, "Astrophysics for Physicists", Cambridge University Press, New York, 2010.*
4. *Krishnaswami K. S., "Understanding cosmic Panorama", New Age International, 2008.*